Ch 6.2

① a) $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$

$A = X \Lambda X^{-1}$

$\lambda = 1, 3$ (upper)

$\begin{bmatrix} 0 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} =$

$2x_2 = 0 \quad \therefore x_2 = 0.$ $\quad x_1 = 1$ (anything)

$\therefore \quad v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\underline{\lambda = 3}$

$\begin{bmatrix} -2 & 2 \\ 0 & 0 \end{bmatrix} \quad -x_1 + x_2 = 0$

$\therefore \quad v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$X = [\vec{v}_1 \quad \vec{v}_2] = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$

$\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$

$$X^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$\therefore X \Lambda X^{-1} =$$

$$A^3 = X \Lambda^3 X^{-1}$$

$$A^{-1} = X \Lambda^{-1} X^{-1}$$

For, $A = \begin{bmatrix} 1 & 1 \\ 3 & 3 \end{bmatrix}$

2) $X = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$; $X^{-1} = \begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$

$$\Lambda = \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$$

$$X \Lambda X^{-1} =$$

③ $A = X \Lambda X^{-1}$ ,

for $A+2I \rightarrow$ eval $= \Lambda + 2I$

eval $= X$

$$\therefore A+2I = X\Lambda X^{-1} + X(2I)X^{-1}$$

$$= X(\Lambda+2I)X^{-1}$$

④ E. vec (cols of X) are Lin. indep

a) A invertible ✗ ($\lambda$ not given. All must be non zero.)

b) A diagonizable ✓ (e. vec. Lind. key condn)

c) X invertible ✓

d) X diagonizable ✗ (e. vec of X needed.)

↓

n linearly indep Eigen vec needed.

⑤ $A = X\Lambda X^{-1}$ ; $X = I$

$\therefore A = \Lambda$ diagonal Mat.

Now,

X → upper triangular ⇒ upper ..

|| lower || || lower ..

Now, $A = X \Lambda X^{-1}$ = True

↓ ↓ ↓

Tri Diag True

(6) $A = \begin{bmatrix} 4 & 0 \\ 1 & 2 \end{bmatrix}$

$\lambda = 4, 2$

$v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

So, $X = \begin{bmatrix} 2c_1 & 0 \\ c_1 & c_2 \end{bmatrix}$

for $A^{-1} \Rightarrow \lambda = \frac{1}{4}, \frac{1}{2}$

eigvec = $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

(7) $x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} + k_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$$X = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$